

Assignment-3 CSE370
Section:22 Spring 25

Question - 1

[5]

Instructor

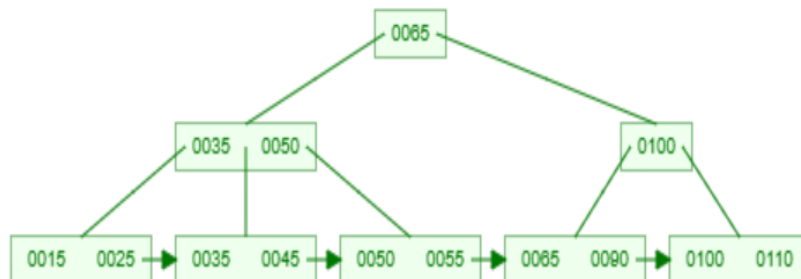
Inst_ID	Surname	Department	Salary	
10101	Srinivasan	Comp. Sci.	65000	
12121	Wu	Finance	90000	
15151	Mozart	Music	40000	
22222	Einstein	Physics	95000	
32343	El Said	History	60000	
33456	Gold	Physics	87000	
45565	Katz	Comp. Sci.	75000	
58583	Califieri	History	62000	
76543	Singh	Finance	80000	
76766	Crick	Biology	72000	
83821	Brandt	Comp. Sci.	92000	
98345	Kim	Elec. Eng.	80000	

- A. Make a hash index on attribute Surname (2nd column). Suppose the number of buckets in the hash index is 5, and each bucket can house 3 index entries at most. Bucket overflow is handled using forward chaining. The values of each character from A-Z (or a-z) are 1-26, respectively. The hash function, $h = (\text{sum of the values of each odd-positioned character in the name}) \% 5$
 Ex: $\text{hash}(\text{Wu}) = (23) \% 5 = 3$
- B. Explain how to search for the search key '**Katz**' in your hash index.

Question - 2

[10]

- A. Given the B+ tree below for $n=4$, delete the following values in the given sequence. After each deletion, show the updated tree. 100, 110, 65, 90, 55, 25, 15



- B. Update the value of the above B+ tree from 50 to 30. After the operation, draw the updated tree.
 C. Insert the value 40 into the above updated B+ tree. After the operation, draw the updated tree.